



Predicting the role of β -GAL genes in bean under abiotic stress and genome-wide characterization of β -GAL gene family members

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Abstract

B-Gals are a subgroup of the glycoside hydrolase (GH) family of enzymes, which possess the Glyco_hydro_35 (GH35) domain. Although studies have been conducted on the β -Gal gene family in numerous plant species, no such research has been conducted on beans. The purpose of this study was to determine the gene expression levels of β -Gal genes in the leaf tissue of *P. vulgaris* under salt and drought stress using quantitative real-time polymerase chain reaction (qRT-PCR) and to perform a comprehensive analysis of β -Gal gene family members using bioinformatics tools. In the bean genome, 25 *Pvul*- β GAL proteins with amino acid numbers ranging from 291 to 1119, molecular weights from 32.94 to 126.56 kDa, and isoelectric points from 5.46 to 9.08 were identified. Both segmental and tandem duplication have occurred in β -Gal genes in the bean genome, and *Pvul*-*BGAL* genes have been subject to negative selection in the evolutionary process. For a deeper comprehension of the evolutionary proximity of *Pvul*-*BGAL* genes, a phylogenetic tree and synteny map were drawn together with *Arabidopsis thaliana* and *Glycine max* β -Gal genes. The expression profiles of β -Gal genes in different tissues of the bean were determined in silico. In addition, the expression profiles of β -Gal genes in the leaves of bean plants subjected to drought and salt stress were analyzed, and the role of β -Gal genes in salt and drought stress was estimated. In this study, the role of β -Gal gene family in abiotic stress response and the characterization of β -Gal genes in beans were determined for the first time and will provide a basis for future functional genomics studies.

Keywords β -galactosidase · *P. vulgaris* · QRT-PCR · GH35 · Glycoside hydrolase (GH)

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Introduction

Global warming, climate change, and environmental pollution cause different abiotic stresses in plants, such as drought, flooding, salinity, and low temperature, as well as unique combinations of biotic stresses, such as plant pathogens, weeds, and insects (Zandalinas et al. 2021). Abiotic stress factors are the largest contributors to agricultural product loss worldwide (Adil and Quraishi 2023). Soil salinity is one of the most important abiotic stress factors in agriculture. Salt tolerance is an important factor for plant productivity and yield continuity (Imran et al. 2021). Drought, an abiotic stress factor, is greatly felt, particularly with increased global warming. Drought and salinity significantly affect plant growth, development, and yield (Muslu et al. 2023). Drought stress in beans reduces plant growth and development as well as the yield and quality of other nutrients and protein content, especially nitrogen (Korkmaz and Durmaz 2017; Yıldırım et al. 2020).

Dry beans have an important place in traditional cuisine culture in Türkiye. In addition, it is also economically important as it is a source of livelihood for rural people. According to FAO data, dry bean cultivation areas were increased by 3.2% from 34.4 million hectares in 2019 to 35.5 million hectares in 2020 (FAOSTAT 2022). World dry bean production has increased by 1.97% in the last 5 years (TurkStat 2022). According to TUIK data, dry beans were grown in 11.82% of the total pulse cultivation area in Türkiye in 2021. The dry bean cultivation area increased by 4.7% compared to the previous year and reached 107 thousand hectares in 2021 (TurkStat 2022). To close the production gap caused by various stress factors, it is necessary to increase the yield per area and to develop new varieties compatible with changing environmental conditions. Identifying and characterizing the genes or gene families involved in plant metabolism during and after stress is the first step to developing plants that are compatible with changing environmental conditions (Muslu et al. 2023). One of these gene families, β -galactosidase (EC 3.2.1.23; β -Gal), is commonly found in all vascular plants. β -Gals are from the subfamilies (GH1, GH2, GH3, GH35, GH42, GH50, and GH59) of the glycoside hydrolase (GH) family (Chandrasekar and van der Hoorn 2016). At the same time, β -galactosidase isozymes in plants are encoded by the β -Gal multigene family (Hou et al. 2021). β -Gal plays an important role in many biological processes, and development, such as seed germination, fruit development, organ elongation, and root development by restructuring the cell wall (Guo et al. 2018). β -Gals control fruit softening during fruit ripening (Husain 2010).

Moreover, β -Gal is also important for cell wall biogenesis and modification during fruit softening. Fruit softening

studies have shown that β -Gal genes in plants preserve fruit firmness during storage (Othman et al. 2011; Guo et al. 2018). β -Gal genes, which play an important role in plant development and environmental stress adaptation, have been identified in many plants such as tomato (*Solanum lycopersicum*), *Arabidopsis thaliana*, rice (*Oryza sativa*), melon (*Cucumis melo* L.), and peach (*Prunus persica* L.). One of the main objectives of this study was to characterize β -Gal genes in bean using various bioinformatics tools. In this study, the expression profiles of β -Gal genes in the leaf tissues of bean plants subjected to salt and drought stress were analyzed using qRT-PCR. These results indicate a role for β -Gal genes in the response to salt and drought stress. The findings of this study will contribute to research on the function of β -Gal genes and provide a new perspective on the role of *Pvul-BGAL* genes in plant stress metabolism. This will also be a preliminary study for future functional genomics studies.

Material-method

Detection of β -GAL genes in *Phaseolus vulgaris*, *Glycine max*, and *Arabidopsis thaliana* genomes

The Pfam Accession Numbers (PF01301, PF02140, PF02449, PF02929, and PF13364) required for the protein sequences of the β -Gal gene family members were retrieved from the Pfam database and screened in the genomes of bean (Schmutz et al. 2014), soybean (Valliyodan et al. 2019), and *Arabidopsis* (Lamesch et al. 2012). β -Gal protein sequences were obtained by keyword searching with these accession numbers in the Phytozome v13 database (Sup. file 1). To identify all possible β -Gal proteins present in the genomes of these plants, both Blastp in Hidden Markov Model (HMM) (<http://www.ebi.ac.uk>) and Phytozome v13 searches were performed in the genomes of these plants. The presence of the β -Gal domain in associated sequences in bean was screened through the HMMER (<http://hmmer.org/>) database. Amino acid number, instability index, stabilization status, theoretical isoelectric point (pI), and molecular weight of β -Gal proteins in bean were determined by the ProtParam tool (<https://web.expasy.org/protparam/>).

Determination of chromosomal location, gene duplications, structure, and conserved motifs of *Pvul-BGAL* genes

The chromosomal locations and sizes of *Pvul-BGAL* genes were determined using Phytozome v13. The positions of the *Pvul-BGAL* genes were tagged with the whole bean chromosomes and visualized using TBTools. Gene duplication events were determined using gene linkage files obtained

from TBTools. The non-synonymous (Ka), synonymous (Ks), and evolutionary forcing (Ka/Ks) between pairs of *Pvul*-BGAL genes were calculated using TBTools (Chen et al. 2023).

To understand the gene structure and organization of *Pvul*-BGAL genes, exon/intron analysis was performed using genomic DNA and CDS sequences stored in databases (Wang et al. 2013). The Gene Structure Display Server v2.0 database was used to obtain details on the coding and non-coding areas of *Pvul*-BGAL genes (Hu et al. 2015).

Additional conserved motifs of the *Pvul*-BGAL amino acid sequences were found using the Multiple Em for Motif Elicitation (MEME) program. (Bailey et al. 2006). The motifs' range in width was set to a minimum of two and a maximum of 50, and the maximum number of motifs was set to 10. The motif regions ranged from 6 to 300. The detected motifs were screened using default settings in the InterPro database (Quevillon et al. 2005).

Sequence alignment and phylogenetic analyses

The neighbor-joining (NJ) model with a bootstrap value of 1000 repetitions was used for phylogenetic analysis. A phylogenetic tree was constructed from the Clustal Omega program using the β -Gal protein sequences of *P. vulgaris*, *A. thaliana*, and *G. max* species (Tamura et al. 2011). A phylogenetic tree was visualized with the Interactive Tree of Life (iTOL) browser (Letunic and Bork 2011).

Comparative mapping between beans and other species

P. vulgaris with *A. thaliana* and *P. vulgaris* with *G. max* genome and transcript sequences were compared using the One Step MCScanX tool. The β -Gal genes of these three plants were scanned in the obtained files, and a multiple synteny map was drawn using the TBtools database tool.

Intracellular localization and promoter analysis of the bean β -GAL

Using the PlantCARE database, cis-acting element analysis was carried out in the 5' upstream region, including around 2 kb DNA fragments of each member of the bean β -Gal gene family (Lescot et al. 2002; Lee et al. 2013). A phenogram was drawn with TBTools (Chen et al. 2023). The WoLF PSORT database was used to predict the subcellular localization of the *Pvul*-BGALs (Horton et al. 2007).

Homology modeling of β -GAL proteins in bean

Protein homology modeling and 3D protein structures were acquired from the Phyre2 database using *Pvul*-BGAL

protein sequences (Kelley et al. 2015). The optimal 3D representation of proteins was identified and shown at the chimera interface by comparing the dependability rates of protein models (Pettersen et al. 2004).

β -GAL protein–protein interactions and gene ontology analysis in *P. vulgaris*

To determine the physical and functional relationships of protein–protein interactions, the STRING (Szklarczyk et al. 2021) database was used and modeled using the Cytoscape (Shannon et al. 2003). For all protein–protein interactions, the data were classified and combined into a reliable score. Gene ontology (GO) analysis, including molecular function, biological process, and cellular component information of *Pvul*-BGAL genes, was performed using TBTools (Chen et al. 2023). GO term analysis was plotted using the SRplot tool (<http://www.bioinformatics.com.cn/SRplot>).

In silico gene expression analyses

To investigate the expression patterns of bean β -Gal genes, Illumina RNA-seq data were acquired from the Sequence Read Archive (SRA) data repository in the NCBI database. Accession numbers for salt (Hiz et al. 2014) and drought (Gregorio Jorge et al. 2020) stress were determined to obtain suitable RNA-seq data. Accession numbers SRR8284480 (drought control), SRR8284481 (drought stress), SRR958469 (salt control), and SRR957668 (salt stress) were used. The reads per million kilobases mapped (RPKM) algorithm was used to normalize the gene expression values (Mortazavi et al. 2008). In addition, transcriptome data of *Pvul*-BGAL genes in different tissues of bean (green mature pods, young pods, stems, roots, nodules, leaves, flowers, and flower buds) were obtained from the Phytozome v13 database as fragments per million kilobases (FPKM). RPKM and FPKM values were log2 transformed, and a heatmap map was obtained using TBTools.

Plant growth and stress application

Commercial bean varieties Serra and Elkoca were used in this study. The seeds of genotypes were surface sterilized for five minutes in a 1% NaOCl solution. Subsequently, the seeds were pre-germinated in pots with perlite. From the germinated seeds, seedlings of a similar size were removed and hydroponically cultivated in pots with 200 mL of a modified 1/10 Hoagland solution. There are many different macro- and micronutrients in Hoagland's solution (Hoagland and Arnon 1950). Bean seedlings were maintained under regulated conditions in a plant development environment with a temperature of 25 °C, 70% humidity, and a light distribution photoperiod of

16/8 h. This allowed the beans to reach the first trifoliolate stage. When the seedlings reached the first trifoliolate stage, they were subjected to salt and dehydration stress. 0 (control) and Hoagland solution with 150 mM NaCl were used. On the ninth day of application, leaf tissue was collected from the bean plants to assess the ion effect of salt stress (Rakhimzhanova et al. 2023). In drought stress, bean plants grown in comparable terms were incubated in a Hoagland solution including 0 (control) and 20% PEG6000 for 24 h (Aygören et al. 2023). At the end of the period, the leaf tissues from the bean plants were immersed in liquid nitrogen and stored at -80°C until the study was completed. The bean varieties utilized in the study were seeded in three replicates per treatment.

qRT-PCR analyses of β -Gal genes in *P. vulgaris* leaves subjected to drought and salt stress

β -GAL primers were constructed using the CDS sequences of *Pvul*-BGAL genes (Table 1). RNA was extracted from drought and salt-stressed bean leaves using the TRIzol® reagent (Invitrogen Life Technologies). Total RNA concentrations were detected at $A_{260/280}$ in Nanodrop and seen in a 1.2% agarose gel. The total RNA quantities were equalized, and cDNA was generated using the SensiFAST™ cDNA Synthesis Kit (Cat No: Bio-65053, UK). *Pvul*-BGAL genes to be employed in qRT PCR analysis were chosen based on transcriptome data. The bean β -Actin gene was employed as the housekeeping gene. qRT-PCR was done in three technical duplicates.

Assessment of data

The expression patterns of the aforementioned genes in abiotic-stressed plants and control samples were normalized to *Pv*- β -Actin expression levels. The data were statistically analyzed using the $2^{-\Delta\Delta\text{CT}}$ technique (Livak and Schmittgen 2001) and two-way ANOVA (multiple comparison test with Bonferroni adjustment). These data were statistically analyzed using GraphPad Prism 9.

Result

Identification and sequence analysis of *Pvul*-BGALs

A keyword search in Phytozome v13 using Pfam accession numbers identified 25 bean *Pvul*-BGAL genes. Table 2 lists the isoelectric points (pI), chromosomal positions, amino acid numbers, and molecular weights (kDa) of the discovered β -Gal genes. The β -Gal protein sequences range in length from 291 (*Pvul*-BGAL-25) to 1119 (*Pvul*-BGAL-24), while the molecular weight (MA) ranges from 32.94 to 126.56 kDa. Furthermore, the isoelectric points of *Pvul*-BGAL proteins range from 5.46 (*Pvul*-BGAL-16) to 9.08 (*Pvul*-BGAL-13). Most *Pvul*-BGAL genes were stable and the instability index ranged from 33.58 (*Pvul*-BGAL-16) to 49.51 (*Pvul*-BGAL-25).

Chromosome position, collinearity, and Ka/Ks analysis of *Pvul*-BGAL genes

In the bean genome, *Pvul*-BGAL genes are distributed on 7 chromosomes (Fig. 1). The Chr09 chromosome of bean has the highest number of *Pvul*-BGAL genes. To analyze the evolutionary status of *Pvul*-BGAL genes, duplication types in the bean genome were studied. As a result, 11 segmentally and 1 tandemly duplicated gene pairs were identified among *Pvul*-BGAL genes.

In gene duplication analysis, *Pvul*-BGAL-11/*Pvul*-BGAL-12 were identified as tandemly duplicated genes, while *Pvul*-BGAL-1/*Pvul*-BGAL-5, *Pvul*-BGAL-1/*Pvul*-BGAL-9, *Pvul*-BGAL-2/*Pvul*-BGAL-5, *Pvul*-BGAL-4/*Pvul*-BGAL-8, *Pvul*-BGAL-4/*Pvul*-BGAL-17, *Pvul*-BGAL-6/*Pvul*-BGAL-23, *Pvul*-BGAL-8/*Pvul*-BGAL-17, *Pvul*-BGAL-8/*Pvul*-BGAL-18, *Pvul*-BGAL-10/*Pvul*-BGAL-20, *Pvul*-BGAL-13/*Pvul*-BGAL-14, and *Pvul*-BGAL-17/*Pvul*-BGAL-18 were segmentally duplicated genes (Fig. 1).

In addition, non-synonymous divergence (Ka) and synonymous divergence (Ks) values, their ratio (Ka/Ks) and segmentally and tandemly duplications of β -Gal genes, and their evolutionary processes were calculated to obtain more detailed information about the evolution of duplicated genes

Table 1 Primer sequences used in qRT-PCR analysis

Gene name	Forward primer	Reverse primer
<i>Pv</i> - β -actin	TGAGCAAGGAGATTACAGCATTGG	CATACTCTGCCTTCGCAATCCAC
<i>Pvul</i> -BGAL-2	GATCTGGTGAGGTTCTGTAAG	ATGCCCGGAACATACTTGAG
<i>Pvul</i> -BGAL-5	TGGTATCTCTAATGTTGCGTGG	CTGACCTTTACCCATTCCCTC
<i>Pvul</i> -BGAL-10	GGAGATGTTTCATCGAGGCTATTC	GCATAATCGGTGGTGTCTT
<i>Pvul</i> -BGAL-13	CGAAGCCACTCACATGGTATAA	ACCCAAGCATAACCCTTTCC
<i>Pvul</i> -BGAL-23	CAGATGCTCCTGACCCAATTA	AAACCATCCAGTCCAGTTCTC

Table 2 Table of information about β -Gal proteins detected in the *P. vulgaris* genome

Gene name	Phytozone id	Chr Loc	Start (bp)	End (bp)	Strand	Aa	MW (kDa)	Pi	Instability	Instability index
<i>Pvul-BGAL-1</i>	Phvul.001G010900	Chr01	830,550	836,614	(-)	739	82.23	8.49	38.89	Stable
<i>Pvul-BGAL-2</i>	Phvul.002G167200	Chr02	32,226,379	32,232,092	(-)	852	94.95	6.88	40.09	Unstable
<i>Pvul-BGAL-3</i>	Phvul.003G050100	Chr03	6,249,845	6,276,305	(-)	912	102.54	6.45	38.42	Stable
<i>Pvul-BGAL-4</i>	Phvul.003G137000	Chr03	33,513,592	33,519,221	(-)	833	92.53	6.25	40.18	Unstable
<i>Pvul-BGAL-5</i>	Phvul.003G258300	Chr03	49,791,140	49,796,926	(-)	847	94.22	6.44	39.92	Stable
<i>Pvul-BGAL-6</i>	Phvul.005G135933	Chr05	37,738,171	37,744,984	(-)	840	91.58	6.48	36.46	Stable
<i>Pvul-BGAL-7</i>	Phvul.005G158700	Chr05	38,723,211	38,734,421	(+)	709	80.18	7.10	36.47	Stable
<i>Pvul-BGAL-8</i>	Phvul.005G160000	Chr05	38,840,181	38,846,204	(-)	837	92.86	6.70	37.66	Stable
<i>Pvul-BGAL-9</i>	Phvul.009G012700	Chr09	2,122,105	2,127,526	(+)	738	82.60	8.64	40.81	Unstable
<i>Pvul-BGAL-10</i>	Phvul.009G119400	Chr09	18,248,056	18,252,334	(+)	845	94.80	7.63	34.71	Stable
<i>Pvul-BGAL-11</i>	Phvul.009G119566	Chr09	18,270,739	18,274,950	(+)	843	94.52	7.14	36.81	Stable
<i>Pvul-BGAL-12</i>	Phvul.009G119632	Chr09	18,282,295	18,286,473	(+)	843	94.49	7.14	35.50	Stable
<i>Pvul-BGAL-13</i>	Phvul.009G127500	Chr09	19,395,194	19,404,067	(+)	735	82.14	9.08	36.10	Stable
<i>Pvul-BGAL-14</i>	Phvul.009G153000	Chr09	22,701,122	22,708,114	(+)	716	80.42	6.00	35.09	Stable
<i>Pvul-BGAL-15</i>	Phvul.009G153100	Chr09	22,714,722	22,724,343	(+)	792	88.22	6.83	35.43	Stable
<i>Pvul-BGAL-16</i>	Phvul.009G225100	Chr09	33,929,463	33,934,776	(-)	706	78.44	5.46	33.58	Stable
<i>Pvul-BGAL-17</i>	Phvul.009G232500	Chr09	34,915,000	34,921,395	(-)	722	79.97	7.56	36.98	Stable
<i>Pvul-BGAL-18</i>	Phvul.010G139300	Chr10	42,158,224	42,164,345	(-)	844	93.60	8.55	38.84	Stable
<i>Pvul-BGAL-19</i>	Phvul.011G075900	Chr11	1,916,814	1,922,776	(-)	817	89.33	7.49	39.77	Stable
<i>Pvul-BGAL-20</i>	Phvul.011G035300	Chr11	3,263,066	3,266,642	(-)	677	75.38	8.31	36.79	Stable
<i>Pvul-BGAL-21</i>	Phvul.011G074300	Chr11	6,929,113	6,935,371	(-)	833	90.87	6.29	40.68	Unstable
<i>Pvul-BGAL-22</i>	Phvul.011G074900	Chr11	6,997,396	7,003,394	(-)	838	91.89	7.51	39.66	Stable
<i>Pvul-BGAL-23</i>	Phvul.011G077600	Chr11	7,136,320	7,143,285	(+)	831	90.39	6.17	36.35	Stable
<i>Pvul-BGAL-24</i>	Phvul.011G206800	Chr11	52,386,468	52,395,605	(+)	1119	126.55	5.57	40.53	Unstable
<i>Pvul-BGAL-25</i>	Phvul.L001654	scaffold_240	436	2959	(-)	291	32.94	7.02	49.51	Unstable

(Table 3). The Ka/Ks ratio for beans ranged from 0.05 to 0.18. In general, a Ka/Ks > 1 means positive selection, Ka/Ks < 1 means negative selection, and Ka/Ks = 1 is known as neutral selection (Li et al. 2009; Khan et al. 2020). As a result, duplicated β -Gal genes in beans were found to undergo purifying selection.

A synteny map was created to understand the evolutionary relationship between beans and *Arabidopsis* and soybeans. In the synteny analysis between *P. vulgaris* and *A. thaliana*, 14 syntenic links were determined, and in the synteny analysis between *P. vulgaris* and *G. max*, 20 syntenic links were determined (Fig. 2).

Intron–exon modeling, conserved motifs, and domain analysis of *Pvul-BGALs*

Gene duplication plays a key role in organismal evolution. Three main mechanisms of exon/intron diversification of multiple gene families are intron/exon gain or loss, exonization or pseudoexonization, and deletion/insertion (Xu et al. 2012; Kasapoğlu et al. 2020). The exon–intron structure of 25 *Pvul-BGAL* genes was analyzed. Many genes have more than one exon/intron in their structure. The number of exons

of the *Pvul-BGAL* genes ranged between 9 and 19, and the number of introns ranged between 8 and 18. Furthermore, *Pvul-BGAL-21* and *Pvul-BGAL-22* genes had similar structural intron lengths and contained 19 exons. The gene with the least number of introns and exons is *Pvul-BGAL-25*, while the genes with the highest number of introns and exons are *Pvul-BGAL-2*, *Pvul-BGAL-3*, *Pvul-BGAL-4*, *Pvul-BGAL-5*, *Pvul-BGAL-6*, *Pvul-BGAL-7*, *Pvul-BGAL-8*, *Pvul-BGAL-18*, *Pvul-BGAL-20*, *Pvul-BGAL-21*, *Pvul-BGAL-22*, and *Pvul-BGAL-23* (Fig. 3).

β -Gal genes have generally structural similarities. As shown in Fig. 3, *Pvul-BGAL-10*, *Pvul-BGAL-11*, and *Pvul-BGAL-12* genes have similar intron–exon distribution with 17 exons. In addition, *Pvul-BGAL-5* and *Pvul-BGAL-9* have similar exon numbers. Thus, it shows that there are similarities in the gene structure of β -Gal genes (Fig. 3).

In the motif analysis performed using the MEME program, a total of 10 different motifs, named Motif1–10, were identified. While most of the *Pvul-BGAL* proteins contain almost all motifs, *Pvul-BGAL-24* protein does not contain any motif. The lowest number of motifs was detected in the *Pvul-BGAL-25* (3) and *Pvul-BGAL-7* (5) proteins (Fig. 4). Motif 3 is present in all β -Gal proteins except *Pvul-BGAL-7*

Fig. 1 Distribution of chromosomal locations of β -Gal genes in *P. vulgaris*, genes shown in pink indicate tandem duplication, and genes shown in blue indicate segmental duplication

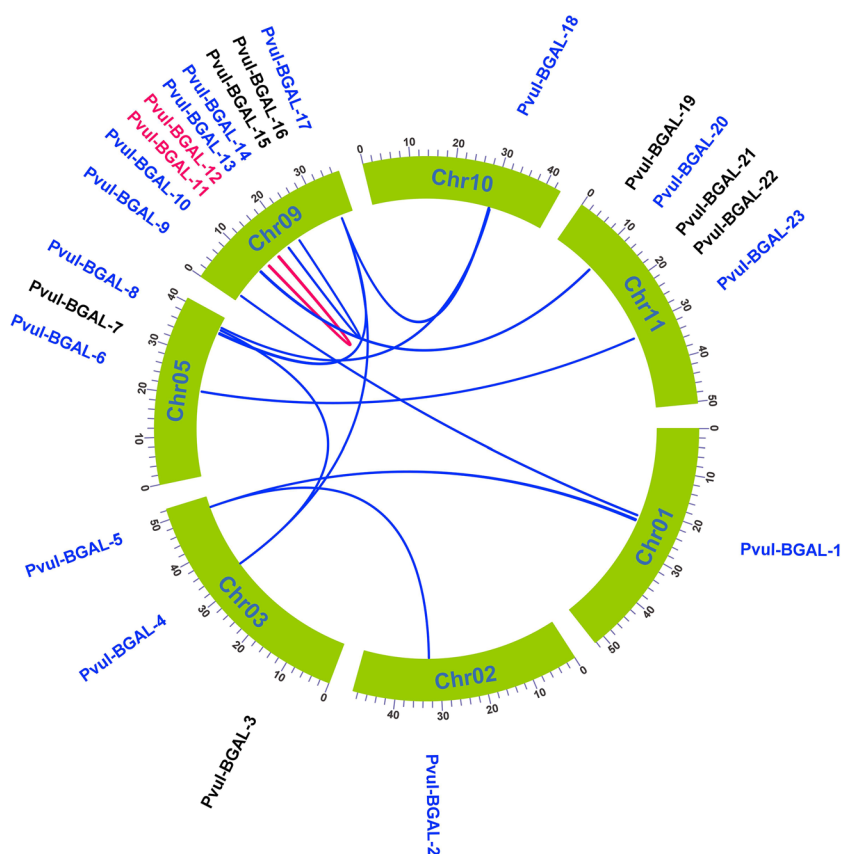


Table 3 Evolutionary status of duplicated *PvuI*-BGAL gene pairs in *P. vulgaris*

Gene 1	Gene 2	Ka	Ks	Ka/Ks	Selective pressure	Duplicate type
<i>PvuI</i> -BGAL-1	<i>PvuI</i> -BGAL-5	0.18	12.59	0.14	Purifying	Segmental
<i>PvuI</i> -BGAL-1	<i>PvuI</i> -BGAL-9	0.10	0.57	0.18	Purifying	Segmental
<i>PvuI</i> -BGAL-2	<i>PvuI</i> -BGAL-5	0.07	0.57	0.13	Purifying	Segmental
<i>PvuI</i> -BGAL-4	<i>PvuI</i> -BGAL-8	0.19	15.87	0.12	Purifying	Segmental
<i>PvuI</i> -BGAL-4	<i>PvuI</i> -BGAL-17	0.09	0.57	0.16	Purifying	Segmental
<i>PvuI</i> -BGAL-6	<i>PvuI</i> -BGAL-23	0.09	0.53	0.17	Purifying	Segmental
<i>PvuI</i> -BGAL-8	<i>PvuI</i> -BGAL-17	0.20	13.35	0.15	Purifying	Segmental
<i>PvuI</i> -BGAL-8	<i>PvuI</i> -BGAL-18	0.073	0.52	0.14	Purifying	Segmental
<i>PvuI</i> -BGAL-10	<i>PvuI</i> -BGAL-20	0.51	NaN	NaN	NaN	Segmental
<i>PvuI</i> -BGAL-13	<i>PvuI</i> -BGAL-14	0.29	19.36	0.15	Purifying	Segmental
<i>PvuI</i> -BGAL-17	<i>PvuI</i> -BGAL-18	0.18	15.00	0.12	Purifying	Segmental
<i>PvuI</i> -BGAL-11	<i>PvuI</i> -BGAL-12	0.00	0.03	0.05	Purifying	Tandem

and PvuI-BGAL-24. Motif 2 is present in all β -Gal proteins except PvuI-BGAL-19 and PvuI-BGAL-24. Motif 1 is present in all β -Gal proteins except PvuI-BGAL-15, PvuI-BGAL-19, and PvuI-BGAL-24. These motifs represent the Glyco_hydro_35 and GHD domains (Fig. 4).

All PvuI-BGAL proteins have G-G active site consensus sequences with two glutamate residues from the GH35 family (Fig. 5). In addition, the best possible matching sequences corresponding to the motifs are shown in Table 4. Using the HMMER database, the Glyco_hydro_35 domain

was identified in *P. vulgaris*, *G. max*, and *A. thaliana* plants. All motifs except motif 10 contained Glyco_hydro_35 and GHD domains.

Cellular localization of *PvuI*-BGAL genes

The cellular locations of *P. vulgaris* β -Gal genes were determined using the WoLF PSORT tool (Table 5). The analysis revealed that PvuI-BGAL proteins are localized in 13 different cellular locations. Bean β -Gal proteins,

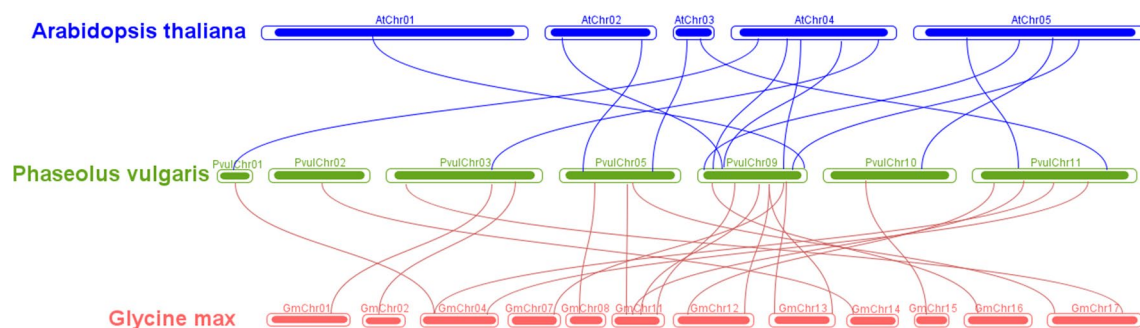


Fig. 2 Genome-wide synteny analysis of *Pvul*-BGAL genes. Comparative map between *P. vulgaris* and *A. thaliana*, *P. vulgaris* chromosomes are shown in green boxes, *A. thaliana* chromosomes in blue

boxes, and *G. max* chromosomes in red boxes (blue lines indicate the link between *A. thaliana*-*P. vulgaris* genomes, and red lines indicate the relationship between *P. vulgaris*-*G. max* genomes)

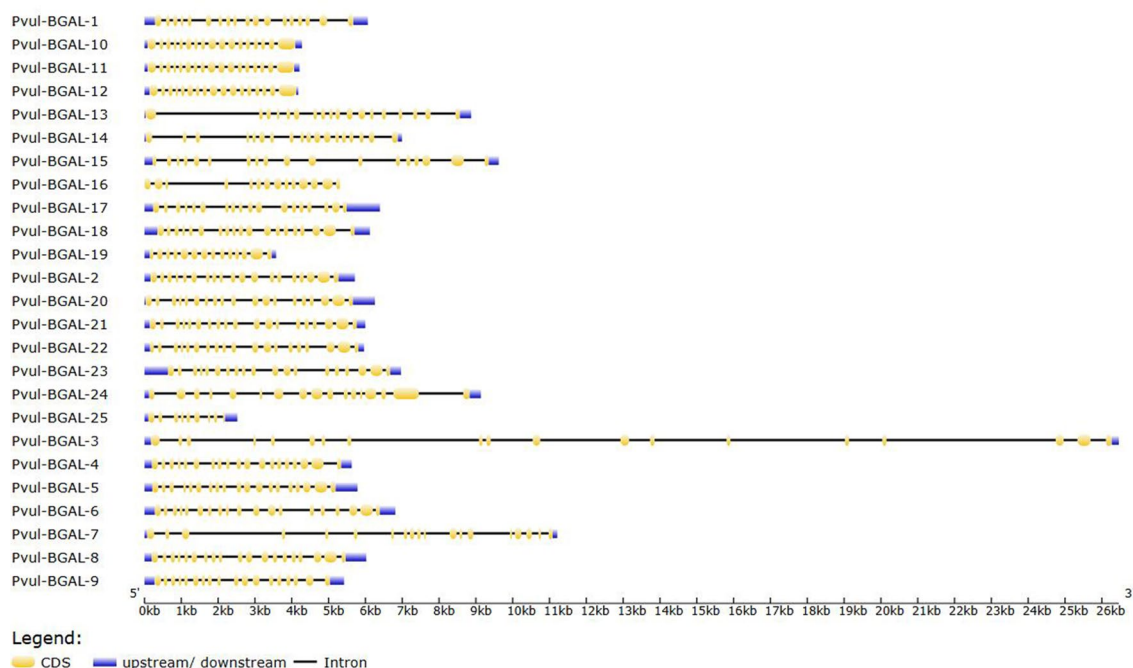


Fig. 3 The number, length, and position of exons and introns in the *Pvul*-BGAL genes are shown. The 5-UTR and 3-UTR in the *Pvul*-BGAL genes are shown in orange boxes. Exons are shown in green

boxes and black lines indicate intron regions. The intron–exon numbers of *Pvul*-BGAL genes are shown on the right side

on the other hand, are primarily found in the chloroplast, vacuole, and mitochondria and are also abundant in the cytoplasm. The localization of some β -Gal genes varied. *Pvul*-BGAL proteins were found to be the least abundant in the Golgi. Interestingly, *Pvul*-BGAL-1 and *Pvul*-BGAL-13 genes had the highest number of cellular locations, while *Pvul*-BGAL-15 and *Pvul*-BGAL-18 had the least number of cellular locations.

Phylogenetic analysis and amino acid sequence alignment of *Pvul*-BGAL transcription factors

To better understand the evolutionary relationship between β -Gal genes in the *G. max*, bean, and *Arabidopsis* genomes, a phylogenetic tree was drawn. It was determined that these genes were grouped into three different groups (A, B, and C) in the phylogenetic tree. Additionally, group C was divided

Fig. 4 Illustration of predicted motif distribution in Pvul-BGAL proteins. 10 different motifs are shown in different colors

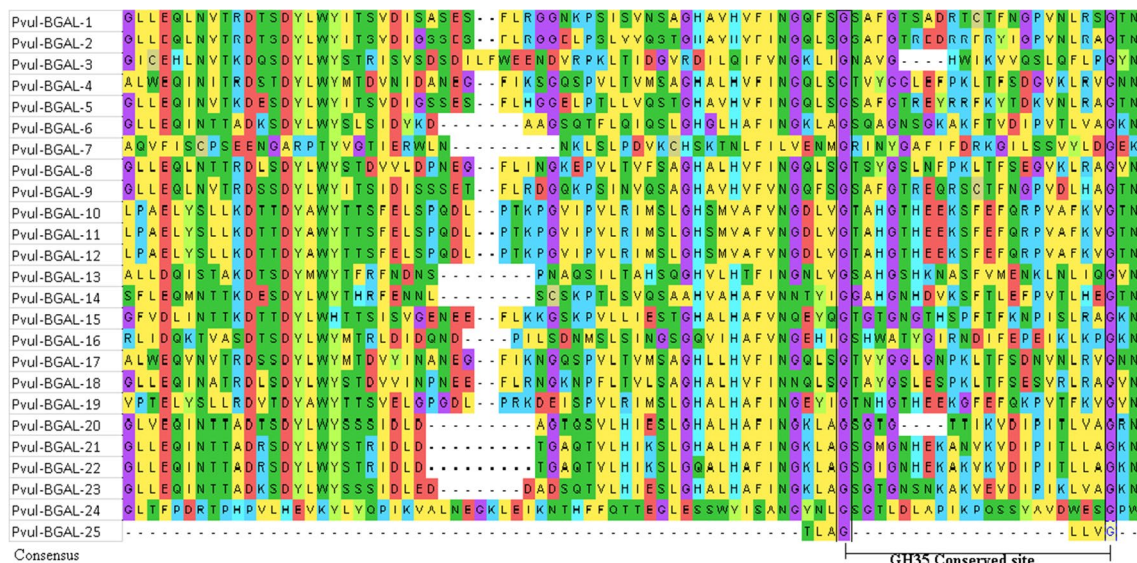
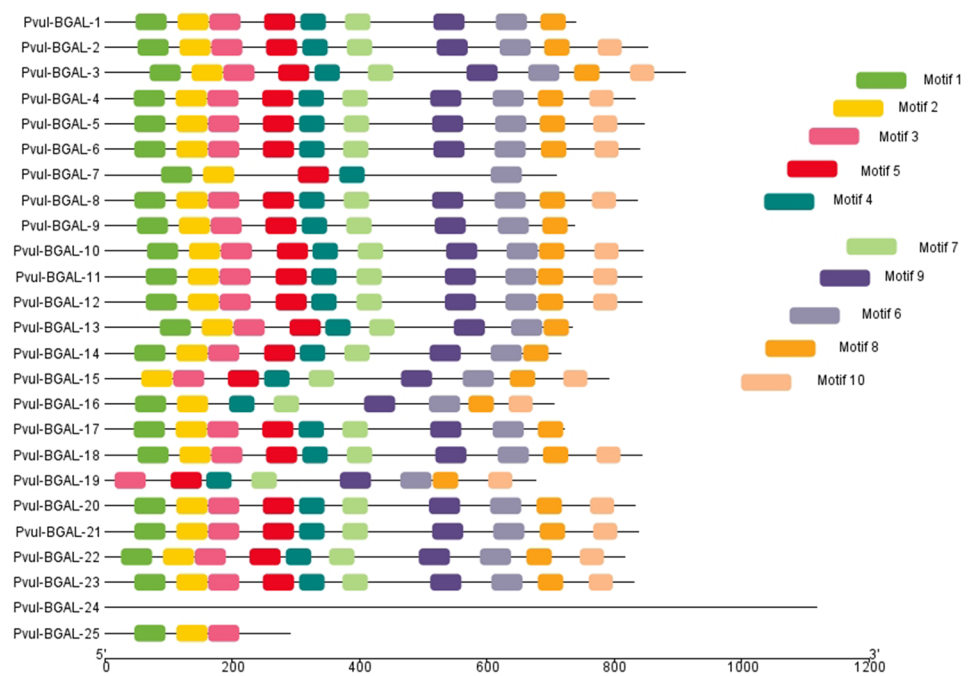


Fig. 5 Multiple sequence alignment of Pvul-BGALs peptides and the GH35 conserved region

into two subgroups (C-I and C-II) (Fig. 6). *G. max* was included in all groups. Among β -Gal proteins, only Pvul-BGAL-10 protein was in group “A” according to the distance matrix. In group C, there were *Pvul-BGAL-11* and *Pvul-BGAL-12*, tandemly duplicated gene pairs. *Pvul-BGAL-6*/*Pvul-B-GAL-23*, a segmentally duplicated gene pair, was in group B. *Pvul-BGAL-1*/*Pvul-BGAL-5*, *Pvul-BGAL-1*/*Pvul-B-GAL-9*, *Pvul-BGAL-4*/*Pvul-BGAL-8*, *Pvul-BGAL-4*/*Pvul-BGAL-17*, and *Pvul-BGAL-8*/*Pvul-BGAL-17*, segmentally gene pairs, were located in group C-I. Group C-II was the

largest group, with 40 members. The group with the fewest members was group A.

Gene structure prediction and 3D modeling

Bean β -Gal protein sequences obtained from the databases were uploaded to the Phyre2 database, and high-probability 3D modeling of the proteins was performed. The 3D predicted structures of *P. vulgaris* β -Gal proteins are shown in Fig. 7.

Table 4 Sequence information of possible motifs in Pvul-BGAL proteins

Motif no	Width	Best possible match	Domain
MOTIF-1	50	IHYPRSTPEMWPDLIQKAKEGGLDVIETYVFWNLHEPQRGQYNFEGRYDL	Glyco_hydro_35, GHD
MOTIF-2	50	RIGPYVCAEWNFGGFPVWLHYIPGIQFRDNEPFKAEMQRFTAKIVDDMMK	Glyco_hydro_35, GHD
MOTIF-3	50	QEKLFASQGGPIILSQIENEYGNIEWAYGEAGKSYVQWAAKMAVSLDTGV	Glyco_hydro_35, GHD
MOTIF-4	41	FITTSYDYDAPLDEYGLQRQPKWGHKLKDLHKAIKLCEEALV	Glyco_hydro_35, GHD
MOTIF-5	50	WTENWTGWTFTEGGPVPHPAEDLAFVARFFQRGGSFVNYYMYHGGTNF	Glyco_hydro_35, GHD
MOTIF-6	50	PKNQPLTWYKTTFDAPEGTDPVALDMTGMGKGQAWVNGQSIGRYWMTYAA	Glyco_hydro_35, GHD
MOTIF-7	41	ANYHTKSDATVTFRNMHYHLPWSISILPDCKNVVFNTAKV	Glyco_hydro_35, GHD
MOTIF-8	41	CGKPTQRWYHVPRSWLKPTDNLLVLFEEEGGNPNQISFVTR	Glyco_hydro_35, GHD
MOTIF-9	50	YGTHERKFTFQRPVTLRAGTNKISLLSVTVGLPNYGAHFETWNAGITGP	Glyco_hydro_35, GHD
MOTIF-10	39	KISSIKFASFGTPQGTGCGNFKHGRCHAHKSYQIVEKQCI	NA

Table 5 Cellular locations of Pvul- β GAL genes

Gene ID	Cellular location
<i>Pvul-BGAL-1</i>	vacu: 3, E.R.: 3, golg: 2, chlo: 1, nucl: 1, cyto: 1, mito: 1, plas: 1, extr: 1
<i>Pvul-BGAL-2</i>	nucl: 6, chlo: 3, cyto: 2, extr: 2, vacu: 1
<i>Pvul-BGAL-3</i>	vacu: 4, chlo: 2, cyto: 2, mito: 2, plas: 2, nucl: 1, extr: 1
<i>Pvul-BGAL-4</i>	chlo: 7, vacu: 4, extr: 2, nucl: 1
<i>Pvul-BGAL-5</i>	chlo: 10, mito: 1, plas: 1, extr: 1, vacu: 1
<i>Pvul-BGAL-6</i>	chlo: 7, extr: 3, nucl: 1, cyto: 1, plas: 1, E.R.: 1
<i>Pvul-BGAL-7</i>	chlo: 9, mito: 3, cyto: 1, plas: 1
<i>Pvul-BGAL-8</i>	vacu: 6, extr: 2, E.R.: 2, golg: 2, chlo: 1, plas: 1
<i>Pvul-BGAL-9</i>	chlo: 7, extr: 3, vacu: 3, nucl: 1
<i>Pvul-BGAL-10</i>	chlo: 9, cyto: 2, mito: 1, plas: 1, extr: 1
<i>Pvul-BGAL-11</i>	chlo: 9, cyto: 2, mito: 1, plas: 1, extr: 1
<i>Pvul-BGAL-12</i>	chlo: 8, cyto: 2, mito: 1, plas: 1, extr: 1, vacu: 1
<i>Pvul-BGAL-13</i>	mito: 3, chlo: 2, nucl: 2, plas: 2, E.R.: 2, cyto: 1, extr: 1, vacu: 1
<i>Pvul-BGAL-14</i>	chlo: 7, extr: 2, E.R.: 2, nucl: 1, cyto: 1, vacu: 1
<i>Pvul-BGAL-15</i>	cysk: 10, nucl: 4
<i>Pvul-BGAL-16</i>	cyto: 5, chlo: 4, plas: 1.5, cysk_plas: 1.5, nucl: 1, mito: 1, pero: 1
<i>Pvul-BGAL-17</i>	nucl: 4, cyto: 3, extr: 3, cysk: 2, chlo: 1, E.R.: 1
<i>Pvul-BGAL-18</i>	chlo: 12, vacu: 2
<i>Pvul-BGAL-19</i>	cyto: 10, cysk: 2.5, cysk_plas: 2, golg: 1
<i>Pvul-BGAL-20</i>	extr: 9, chlo: 4, vacu: 1
<i>Pvul-BGAL-21</i>	chlo: 9, vacu: 2, nucl: 1, plas: 1, E.R.: 1
<i>Pvul-BGAL-22</i>	mito: 8, chlo: 4, nucl: 1, cyto: 1
<i>Pvul-BGAL-23</i>	vacu: 5, chlo: 2, extr: 2, golg: 2, nucl: 1, plas: 1, E.R.: 1
<i>Pvul-BGAL-24</i>	chlo: 5, nucl: 3, cyto: 2, vacu: 2, cysk: 1.5, cysk_plas: 1.5
<i>Pvul-BGAL-25</i>	vacu: 4, chlo: 3, extr: 3, golg: 2, nucl: 1, plas: 1

In the 3D modeling analysis, while the confidence ratio of all Pvul-BGAL proteins was 100%, differences were observed in coverage ratios. The highest coverage rate was seen in Pvul-BGAL-17. The lowest coverage rate was seen in Pvul-BGAL-25. In addition, three types of folding were observed: alpha-helix, beta-strand, and TM-helix (Fig. 7). When the distribution of folding types of proteins was analyzed, it was determined that the alpha-helix (29%) structure was more dominant in Pvul-BGAL-25 compared to other

proteins. Beta-strand structure was found to be dominant in Pvul-BGAL-24 compared to other proteins. In addition, the TM helix, the most common secondary structure found in the transmembrane segments of membrane proteins, was observed in Pvul-BGAL-3, Pvul-BGAL-6, Pvul-BGAL-7, Pvul-BGAL-9, Pvul-BGAL-10, Pvul-BGAL-11, Pvul-BGAL-12, Pvul-BGAL-13, Pvul-BGAL-16, Pvul-BGAL-18, Pvul-BGAL-20, Pvul-BGAL-21, and Pvul-BGAL-24.

Fig. 6 Phylogenetic tree drawn with β -Gal proteins of *Phaseolus vulgaris*, *Glycine max*, and *Arabidopsis thaliana* plants

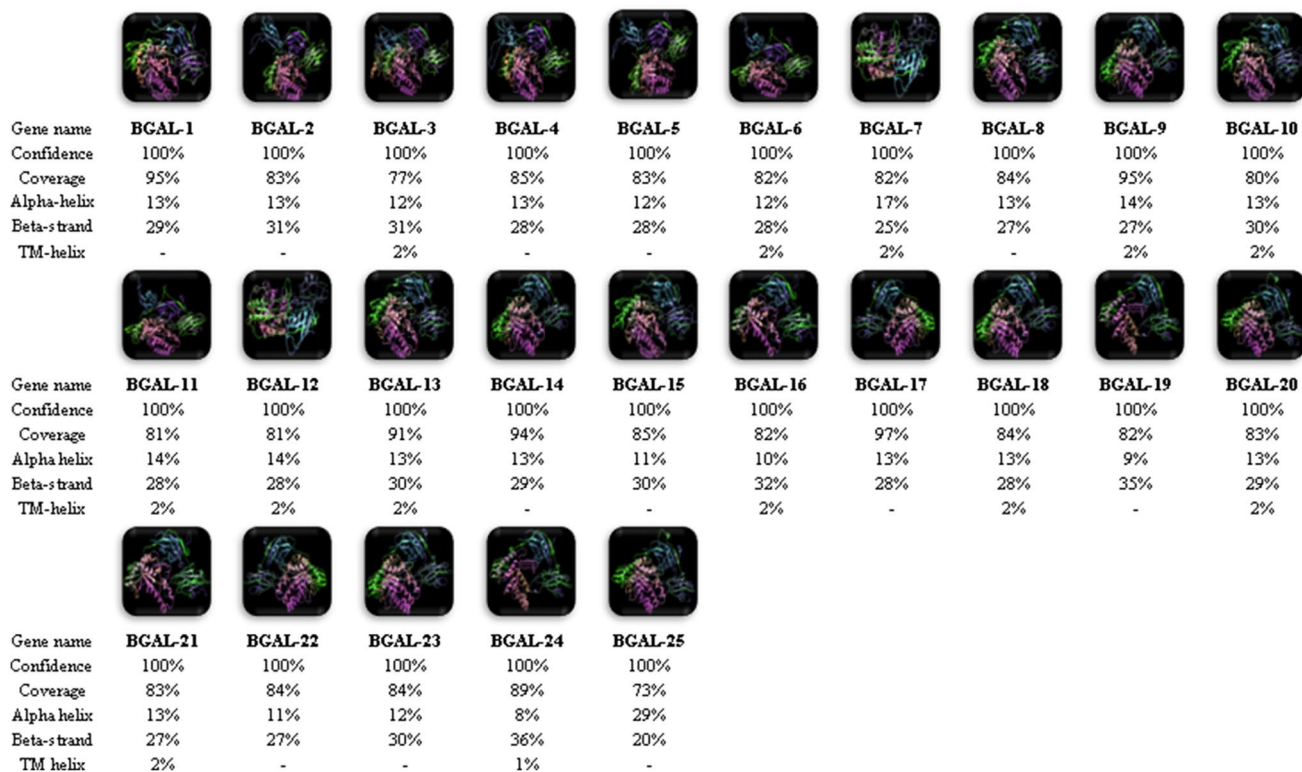


Fig. 7 3D homology modeling of *P. vulgaris* β -Gal genes

Protein–protein interactions and gene ontology identification of *P. vulgaris* β -Gal genes

The ppi analysis using the STRING online interface showing the intercommunication of Pvul-BGAL proteins is shown in Fig. 8. Pvul-BGAL-7 and Pvul-BGAL-24 proteins interact with all other β -Gal proteins. Except for Pvul-BGAL-7 and Pvul-BGAL-24 proteins, other proteins interact in binary pairs. Furthermore, the interactions between Pvul-BGAL proteins and Aldosa 1-epimerase enzymes were detected. In particular, Pvul-BGAL-7 showed a common network interaction on both sides (Fig. 8).

According to the gene ontology results, the biological processes in which Pvul-BGAL proteins are involved are the beta-galactosidase complex, the catalytic complex, and the protein-containing complex. The Pvul-BGAL proteins participate in a variety of molecular functions, which include hydrolase activity, hydrolase activity acting on glycosyl bonds, carbohydrate binding, beta-galactosidase activity, galactosidase activity, and catalytic activity. Its cellular locations are the carbohydrate metabolic process,

the primary metabolic process, and the organic matter metabolic process (Fig. 9).

In addition, protein-specific GOs were identified by GO terms analysis. GO terms GO:0005975, GO:0030246, and GO:0004553 were found in all Pvul-BGAL proteins, while GO:0009341 and GO:0004565 were found in only five proteins (Fig. 10).

Cis-element analysis of *Pvul*-BGAL genes

Cis-acting elements are specific motifs present in the promoter region of each gene that regulates gene transcription. A total of 84 different cis-acting elements were discovered using the PlantCARE program for the characterization of cis-elements of β -Gal genes. Many different cis-acting elements were identified in β -Gal genes, including light-sensitive elements, environmental stress-related elements, plant growth-related elements, promoter-related elements, and plant hormone-responsive elements. All *Pvul*-BGALs contain the promoter elements CAAT-box and TATA-box. *Pvul*-BGAL-1, *Pvul*-BGAL-2, and *Pvul*-BGAL-3 also contain a MYC motif involved in drought. Light response elements

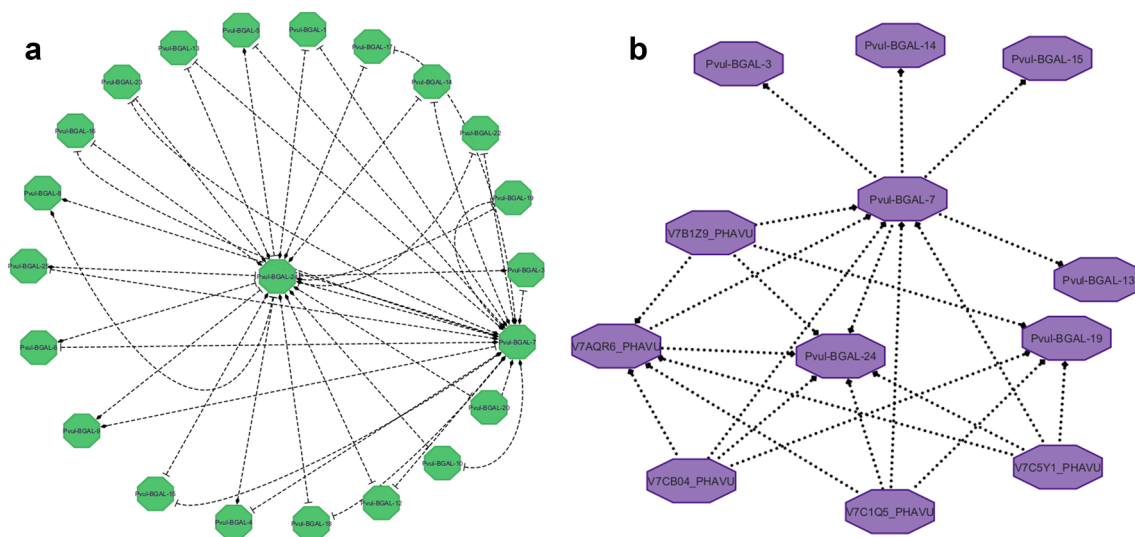


Fig. 8 a, b Pvul-BGAL protein–protein interactions. (In the ppi analysis using STRING, the interaction of the Pvul-BGAL proteins is shown in green on the left and the interaction with 5 different proteins is shown in purple on the left)

Fig. 9 Gene ontology analysis including biological process, molecular function, and localization in which Pvul-BGAL proteins are located (plotted using $-\log_{10}(p\text{-value})$ confidence interval in TBTool)

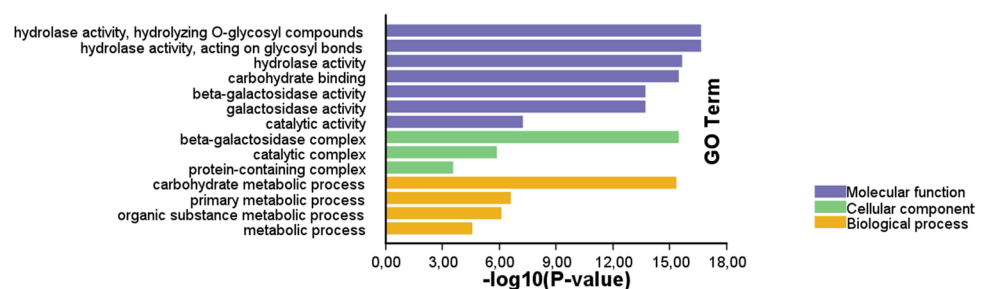
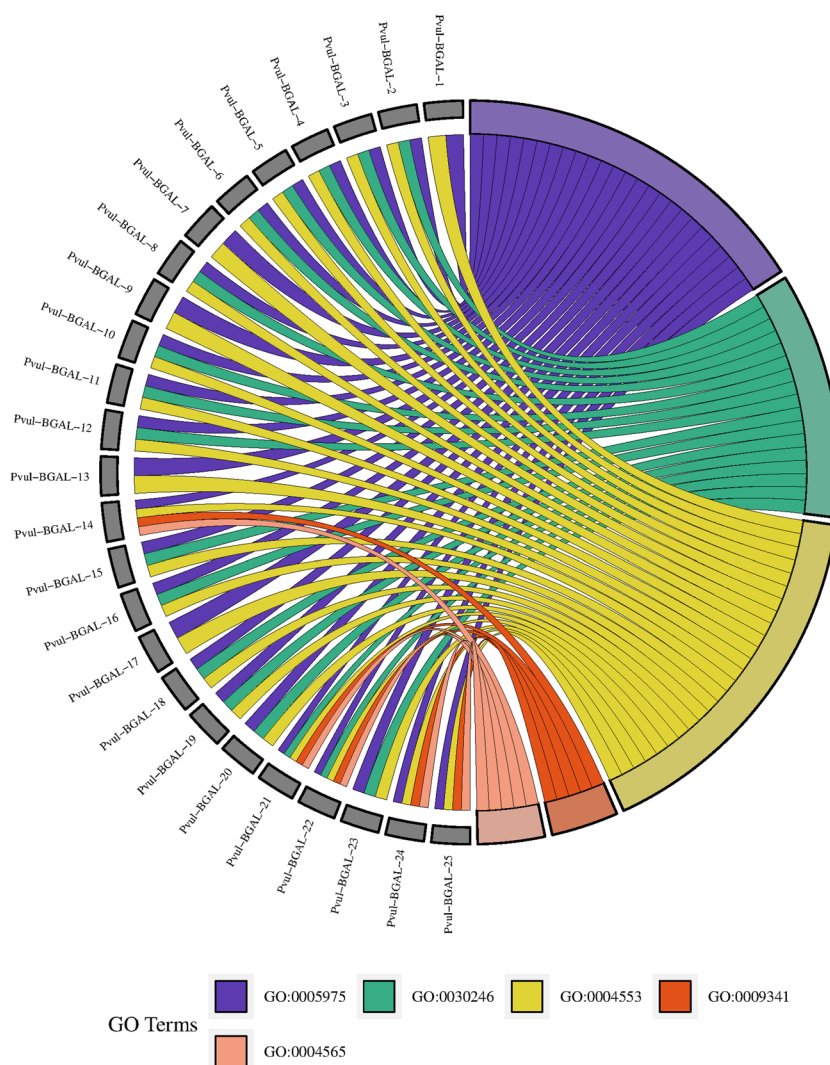


Fig. 10 Analysis of GO Terms (GO:0005975, GO:0030246, GO:0004553, GO:0009341, and GO:0004565) in *Pvul*-BGAL proteins



that may be essential for plant growth and development were also found. In addition, ERE, ARE, MYB, W-box, and TC-rich repeats cis-elements associated with hormones involved in stress were also found in some *Pvul*-BGAL genes (Fig. 11).

The expression pattern of *Pvul*-BGAL genes under abiotic stresses and selected tissue

The drought and salinity expression profile of β -Gal genes of *Pvul*-BGAL genes under abiotic stress factors of the *P. vulgaris* plant was determined (Fig. 12). In drought and salt stress, *Pvul*-BGAL-17 and *Pvul*-BGAL-18 genes were expressed more than other genes. The least expressed genes were identified as *Pvul*-BGAL-10 and *Pvul*-BGAL-11.

As a result, it has been shown that there are significant changes in the expression levels of *Pvul*-BGAL genes under salinity stress. The most significant reduction in expression level under salt stress has been seen in the

Pvul-BGAL-23 gene, while the most significant increase has been seen in the *Pvul*-BGAL-17 and *Pvul*-BGAL-20 genes. In drought stress, increases and decreases in the expression levels of *Pvul*-BGAL genes were determined. The most significant increase in expression levels was observed in *Pvul*-BGAL-18, while the most significant decrease was observed in *Pvul*-BGAL-10 and *Pvul*-BGAL-25 genes.

Expression profiles of β -Gal genes in bean flowers, green mature pods, leaves, nodules, young trifoliates, stems, young pods, flower buds, and root tissues were examined. The highest gene expression in young trifoliolate tissue was seen in *Pvul*-BGAL-23. The most expressed genes in flower and flower bud tissues are *Pvul*-BGAL-10, *Pvul*-BGAL-11, and *Pvul*-BGAL-12. The highest expression level in leaves and nodule tissues was seen in *Pvul*-BGAL-14. *Pvul*-BGAL-13 was also seen in green mature pod tissue. Expression levels of *Pvul*-BGAL genes in root and stem tissues are moderate (Fig. 13).

Fig. 11 Cis-element prediction of *Pvul-BGAL* genes (WRKY, MYC, and as-1 are associated with drought stress, and ABRE and MBS cis-elements are all associated with salt stress)

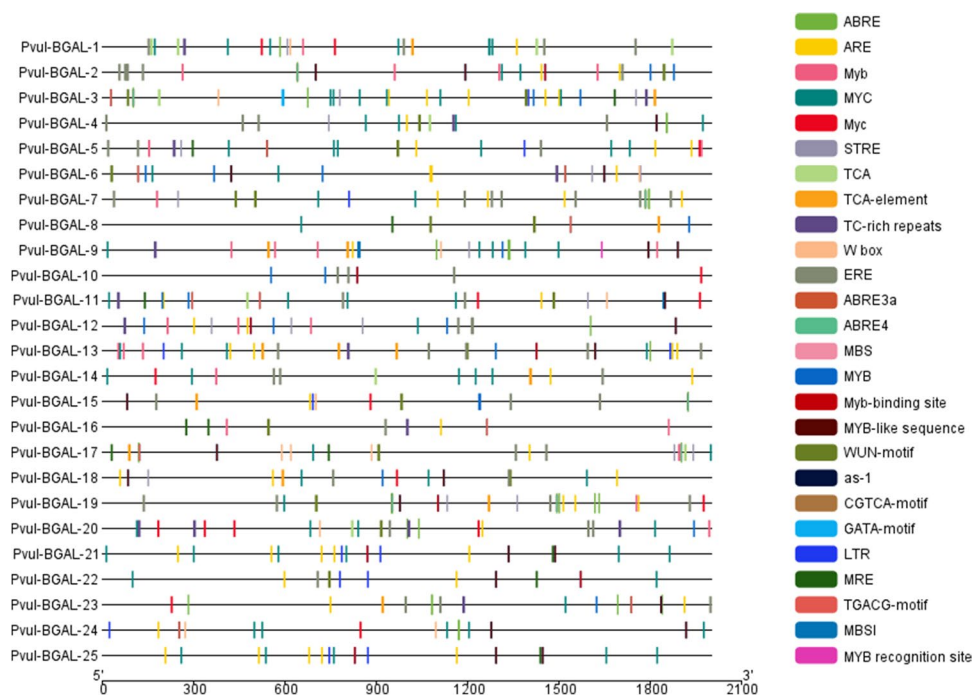
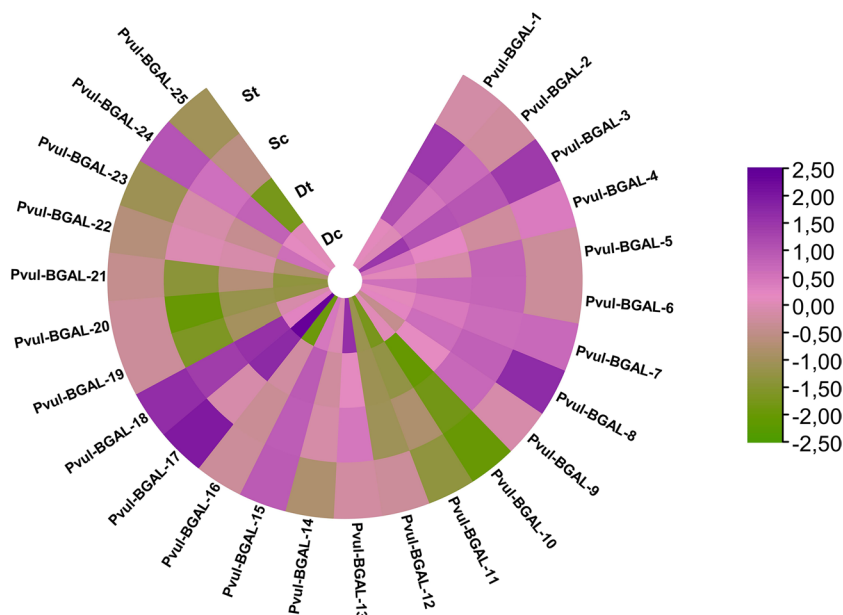


Fig. 12 Expression analysis of members of the β -Gal gene family in *P. vulgaris* under abiotic stress conditions (heat-map) The colors indicate the expression level of the genes. The expression level increases from green to purple. Pink or colors close to pink indicate that the expression level does not change significantly. According to the RPKM values, the stronger the expansion difference between the two colors, the more the expression level of the gene changes



qRT-PCR analyses of β -Gal genes in bean leaves subjected to drought and salt stress

qRT-PCR analysis was performed to examine the effects of salt and drought stress on gene expression in two different bean cultivars (sensitive and resistant). Elkoca-05 cultivar is more resistant than Serra cultivar (Aygören et al. 2023). Five different genes showing different expression profiles in both leaf tissue and stress conditions were preferred. *Pvul-BGAL-2*, *Pvul-BGAL-5*, *Pvul-BGAL-10*,

Pvul-BGAL-13, and *Pvul-BGAL-23* primers were used for qRT-PCR analysis. To induce drought stress, 20% PEG6000 was applied for 24 h. No significant changes were observed in the expression of *Pvul-BGAL-2*, *Pvul-BGAL-5*, and *Pvul-BGAL-23* in both cultivars compared to the control (0% PEG6000). There was a significant decrease in the expression level of *Pvul-BGAL-10* in Elkoca-05 and Serra cultivars, but a significant increase in the expression level of *Pvul-BGAL-13* was observed only in Elkoca-05 cultivar (Fig. 14).

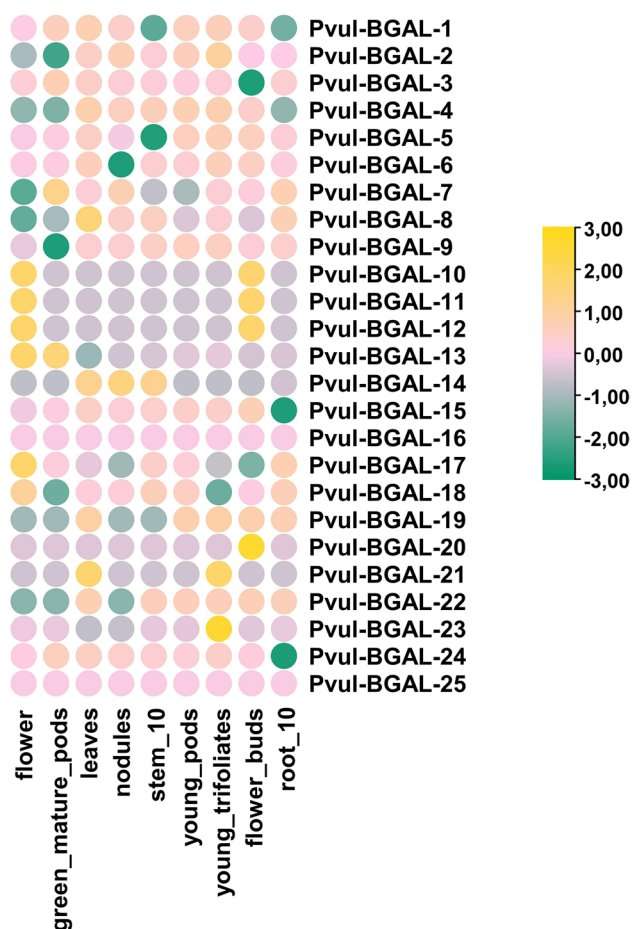


Fig. 13 Expression analysis of β -Gal gene family members in tissues using transcriptome data of *P. vulgaris* plant. Colors show the expression level of *PvuL-BGAL* genes in tissues. The expression level increases from green to yellow. According to the FPKM values, the stronger the expansion difference between the two colors, the more the expression level of the gene changes

No significant change has been seen in the expression level of only *PvuL-BGAL-2* in Elkoca-05 cultivar compared to the control (0 mM NaCl) under salt stress (150 mM NaCl). A significant increase in the expression levels of other *PvuL-BGAL* genes was observed in Elkoca-05 cultivar. In Serra cultivar, a significant increase has been seen only in the expression level of *PvuL-BGAL-2*. No significant changes have been seen in the expression levels of other *PvuL-BGAL* genes in Serra cultivar (Fig. 15). Salt stress treatment resulted in a significant increase in the expression levels of *PvuL-BGAL* genes in general. In both drought and salt stress, a significant change was observed in the expression level of *PvuL-BGAL-10* gene in Elkoca-05 cultivar.

Discussion

β -Galactosidases are involved in plant growth, development, and senescence (Buckeridge and Reid 1994; Martín et al. 2009). The β -Gal gene family, an enzyme that can directly cleave β -(1 $\rightarrow$ 4) galactan bonds (Tateishi 2008), was found to be 25 in bean (*Phaseolus vulgaris*) in this study, 17 in peach (*Prunus persica* (L.) Batsch) (Guo et al. 2018), 21 in melon (*Cucumis melo* L.) (Pan et al. 2022), and 21 in sweet potato (*Ipomoea batatas* (L.) Lam) in other studies. In addition, 17 β -Gal gene family members have been reported in *Arabidopsis* and tomato (*Solanum lycopersicum*) genomes, which have similar protein homology (Gantulga et al. 2009).

The active site consensus sequence GGP-[LIVM]-xQx-ENE-[FY] (PF01301) is common to all GH35 β -Gals and contains the Glu residue that acts as a proton donor during hydrolysis (Chandrasekar and van der Hoorn 2016). In our study, the same sequence and glutamate residue are present in the two active consensus sequences. *PvuL-BGALs* contain only the Glyco_hydro_35 domain, whereas *Ib* β -Gals from the sweet potato study contain an additional Gal_Lectin domain (Hou et al. 2021). In our study, 10 conserved motifs were analyzed, and the GH35 conserved region was identified. In addition, GHD and Gal_Lectin domains are present in melon (*Cucumis melo*) plants besides the GH35 domain but not in our study (Pan et al. 2022). In addition, it is noteworthy that almost all β -Gal genes contain motifs in motif analysis, while *PvuL-BGAL-24* does not contain any motifs. In previous motif analyses, there are β -Gal genes that contain almost no motifs. In *Cucumis melo*, *CmBGAL12* was reported to contain only motif 1 out of ten motifs (Pan et al. 2022). *PvuL-BGAL* proteins, which show the same branching pattern in the phylogenetic tree, have a common motif arrangement. In our study, the phylogenetic tree consists of three different groups. *PvuL-BGAL-11/PvuL-BGAL-12*, which are tandem duplicated genes, are located in group C, and the highest number of members belongs to group C. Unlike our study, the phylogenetic tree in sweet potato and melon studies consists of seven groups, and *Brassica campestris* consists of four groups. (Liu et al. 2013; Hou et al. 2021; Pan et al. 2022). This shows that phylogenetic trees constructed with many different plants may exhibit different branching patterns.

In protein secondary structure analysis, the highest α -helix ratio of *PvuL-BGAL* genes is 29%, and the lowest α -helix ratio is 8%. The highest beta strand helix ratio is 36%, and the lowest beta strand helix ratio is 20%. A study in melon shows that melon *BGAL* gene family members have the highest α -helix ratio of 25.07% and the lowest ratio of 20.98% (Pan et al. 2022). There is often an

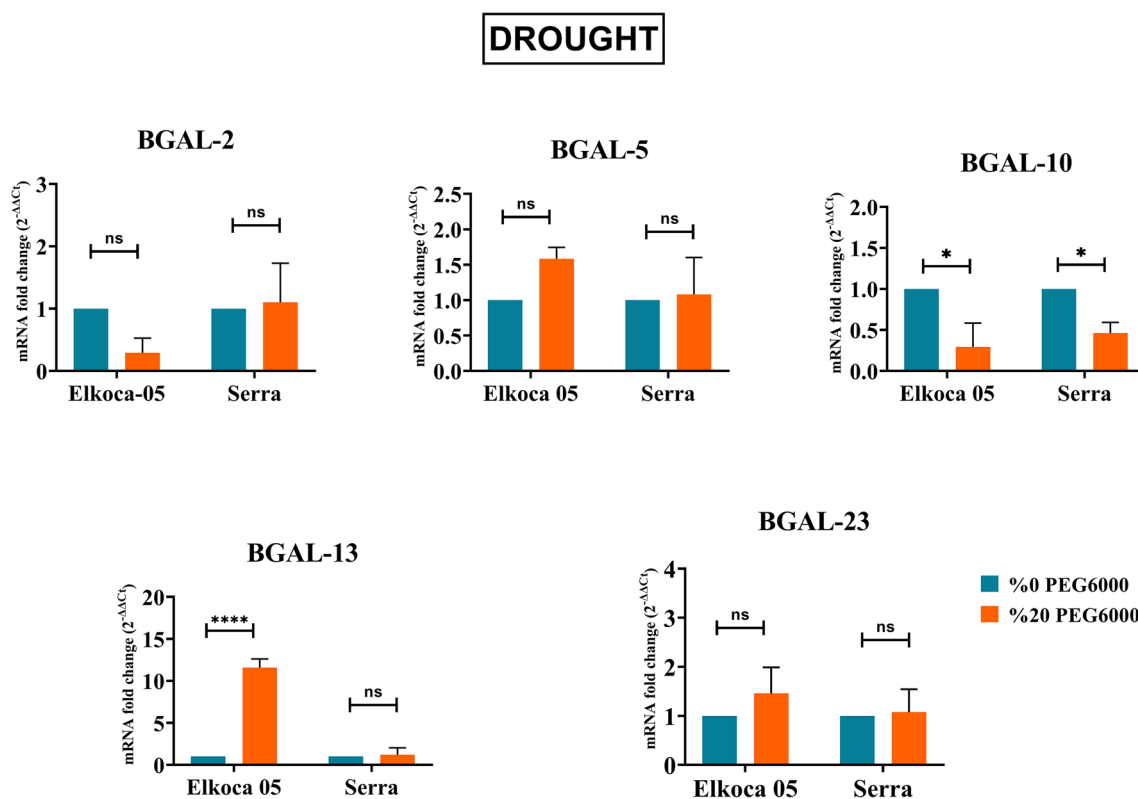


Fig. 14 Expression level plots of *Pvul-BGAL* genes in Elkoca-05 and Serra cultivars under drought stress (20% PEG6000). The Dunnett test revealed significant differences between the treatments and the control (* $p < 0.05$, ** $p < 0.01$, and ns, non-significant)

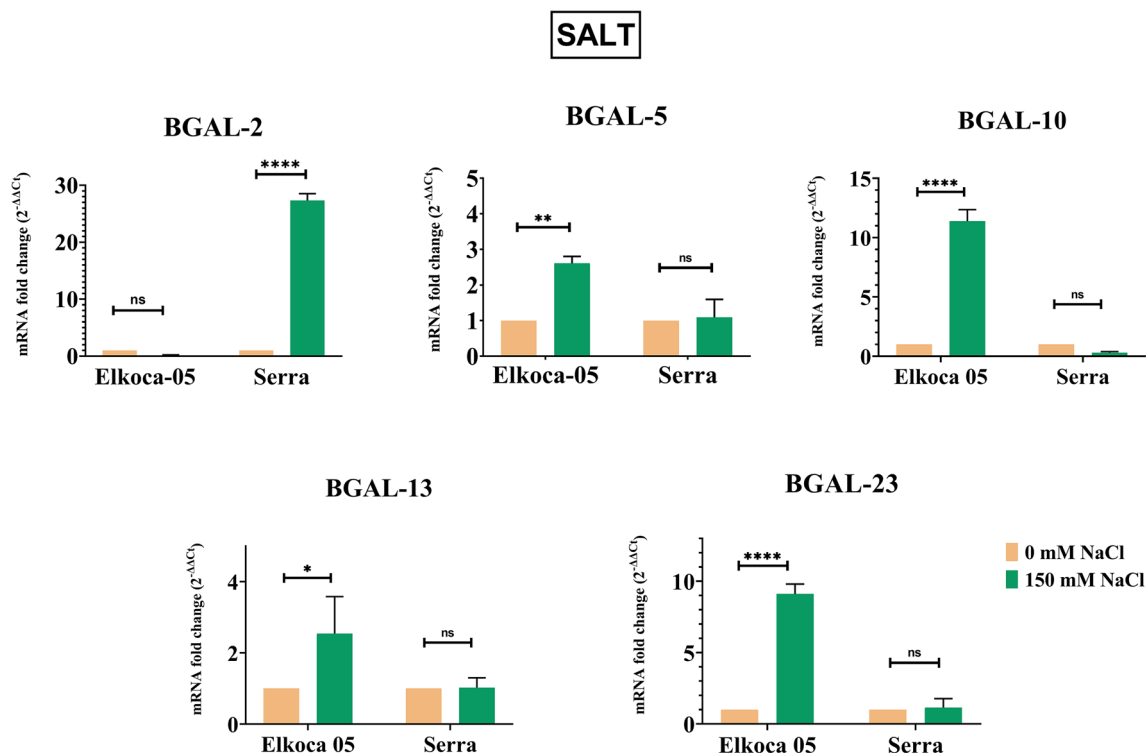


Fig. 15 Expression level plots of *Pvul-BGAL* genes in Elkoca-05 and Serra cultivars under salt stress (150 mM NaCl). The Dunnett test revealed significant differences between the treatments and the control (* $p < 0.05$, ** $p < 0.01$, and ns, non-significant)

interaction between extended family members. *Pvul-BGAL* genes also interact with each other. In particular, *Pvul-BGAL-7*, *Pvul-BGAL-24*, and *Pvul-BGAL-19* proteins have been associated with 5 aldose 1-epimerase enzymes. Aldose 1-epimerase protein is a key enzyme, especially in carbohydrate metabolism (Heese-Peck and Raikhel 1998). *Pvul-BGAL* proteins are also involved in the carbohydrate metabolic process and carbohydrate binding. In addition, GO:0004565, which is the most abundant in molecular functions, is involved in beta-galactosidase activity.

Pvul-BGAL-24 is the *Pvul-BGAL* gene with the highest protein interaction and contains only the G35H conserved domain in its structure. In addition, *Pvul-BGAL-24* exhibited a very different folding pattern from other *Pvul-BGAL* genes in terms of protein structure. Meanwhile, such a member was also discovered in sweet potato (*Ibbgal17*) and melon (*CmBgal12*) (Hou et al. 2021; Pan et al. 2022), indicating that it is a highly conserved member type in plant BGAL families.

Bean β -Gal genes have been subjected to purifying selection during the evolutionary process and exhibited 12 different duplications, 11 segmental and 1 tandem. Gene duplication events are the most important factor in the expansion and evolution of gene families (Holland et al. 1994). Purifying selection, also known as background selection, reduces genetic diversity and shapes it in the natural population (Cvijović et al. 2018). Consequently, it can be suggested that the genetic diversity of *Pvul-BGAL* genes has decreased as a result of purifying selection. It shows that *Pvul-BGAL* genes have been preserved in the evolutionary process. Duplicated gene pairs showed similar structure in intron/exon analysis and motif analysis and similar branching in the phylogenetic tree. *Pvul-BGAL-11* and *Pvul-BGAL-12*, which have similar intron/exons, are in the same group in the phylogenetic tree. Genes with similar introns/exons in *C. melo* and *B. campestris* showed similar branching in the phylogenetic tree (Liu et al. 2013; Pan et al. 2022). In contrast to *Pvul-BGAL* genes, some β -Gal genes of *C. melo* and *P. persica* plants were reported to be intronless (Guo et al. 2018; Pan et al. 2022). Intron regions have an important role in the evolutionary process of genes (Poverennaya and Roytberg 2020).

Most of the *Pvul-BGAL* genes are almost the same size. However, although *Pvul-BGAL-3* is quite long compared to other genes, it contains the same number of exons as other genes. The size of a gene is directly proportional to the length of the transcript it encodes, the size of the protein, and the number of introns (Lopes et al. 2021). But it is incorrect to say that the larger the size of a gene, the more functions it will have. In the gene ontology analysis, the function of *Pvul-BGAL-3* is the same as other genes. In addition, recent studies have concluded that introns effective at the gene size are not completely dysfunctional and play a role in regulating gene expression (Dwyer et al. 2021). Accordingly,

the *Pvul-BGAL-3* gene may differ from other BGAL genes in the regulation of gene expression.

Chromosomal position analysis showed that *Pvul-BGAL* genes are distributed on 7 chromosomes of bean (Chr01, Chr02, Chr03, Chr05, Chr09, Chr10, and Chr11). *Pvul-BGAL* genes are most concentrated on Chr09, suggesting that genes clustered on the same chromosome evolve faster (Zheng et al. 2024). In melon and peach, β -Gal genes are not evenly distributed on chromosomes (Guo et al. 2018; Pan et al. 2022). As a result of synteny analysis between β -Gal genes of *P. vulgaris* and other selected plants, β -Gal genes of *G. max* showed more orthology with bean compared to β -Gal genes of *Arabidopsis*. Closely related species show similar gene organization (Lewis 2005; Restrepo-Montoya et al. 2021). At the same time, synteny analysis with orthologous genes allows a better understanding of genome structure and mechanism of gene function (Wang et al. 2015; İlhan et al. 2018).

Pvul-BGAL proteins are located in different cellular localizations but are concentrated in the chloroplast, vacuole, and mitochondria. In melon, β -Gal proteins are more localized in the cell wall (Pan et al. 2022). In *Arabidopsis* and apple, β -Gal proteins are concentrated in intracellular parts (Yang et al. 2018). This confirms that β -Gal proteins are involved in cell wall metabolism (Moneo-Sánchez et al. 2016). The presence of β -Gal proteins in different cellular locations suggests that they are involved in many different cellular functions (Moneo-Sánchez et al. 2016; Pan et al. 2022).

In response to biotic and abiotic stress, gene expression is regulated by specific cis-regulators in the promoter region, and transcription of some genes is stopped or increased (Kasapoglu et al. 2022). In bean, 84 different cis-acting elements have been identified in the promoter region of β -Gal genes. These were associated with plant hormone response, stress response, and light response. TCA-element, TGACG-motif, ABRE, and MBS cis-elements are all related to salt stress, while MYB, ARE, WRKY, MYC, and as-1 are related to drought stress (Yamniuk and Vogel 2004; Saediazar et al. 2014; Yuan et al. 2021). The MYC motif and MYB, which are drought stress elements, are highest in *Pvul-BGAL-3* and *Pvul-BGAL-4*. This suggests that many of the *Pvul-BGAL* genes are involved in the stress response in addition to fruit softening. Furthermore, all of the *CmBGAL* genes contain the ERE cis-element that responds to ethylene signaling (Pan et al. 2022). In our study, *Pvul-BGAL-7* contains the most ERE cis-element. Ethylene, abscisic acid, and methyl jasmonate signaling have been suggested to be involved in the regulation of transcription of β -Gals, resulting in fruit softening (Guo et al. 2018; Pan et al. 2022).

Various stress factors, especially abiotic stresses, cause changes in the expression levels of β -Gals in plants. In our study, changes in the expression levels of *Pvul-BGAL*

genes were observed under salt and drought stress. In particular, under salt stress, the expression levels of *Pvul-BGAL-17* and *Pvul-BGAL-20* genes increased. In addition, under drought stress, qRT-PCR analysis showed a significant increase in the expression level of *Pvul-BGAL-10* and a significant decrease in the expression level of *Pvul-BGAL-13*. Most *Ib β -Gal* genes in sweet potato have been reported to be associated with salt stress, drought stress, ABA treatment, and pathogen infection (Hou et al. 2021). Expression profiles of β -Gal genes were analyzed in different tissues, and a significant increase was observed in flower tissue of bean. In other tissues, a decrease in expression level was observed. It is known that most of the *CmBGALs* are highly expressed in melon, especially in shoot and stem tissues, whereas in *Arabidopsis*, they are more highly expressed in leaf, root, and flower tissues (Ahn et al. 2007). In addition, it was concluded that it is highly expressed in leaf, stem, and root tissues of tomato (Smith and Gross 2000). As a result of the analysis, it can be said that *Pvul-BGAL* genes are associated with both cell metabolism and drought and salt stress. As a result of the in silico and qPCR analyses obtained, in future functional and breeding studies, *Pvul-BGAL2*, *Pvul-BGAL10*, *Pvul-BGAL17*, and *Pvul-BGAL23* genes, which are thought to play a role in stress response and can be expressed in almost all tissues, can be studied.

Conclusions

As a result of genome-wide analysis of the β -Gal gene family in *P. vulgaris*, 25 *Pvul-BGAL* genes were identified, distributed on seven chromosomes, clustered in three classes, and with significant evolutionary differences. It has also been determined that segmental and tandem duplication events play an important role in the expansion and evolutionary process of the *Pvul-BGAL* family. In the results of in silico analyses, tissue-specific expression profiles of β -Gal genes and their expression levels under various stresses differ in the growth and development of *P. vulgaris*. Quantitative qRT-PCR analyses of five selected *Pvul-BGAL* genes were performed, and accordingly, the *Pvul-BGAL-10* gene is thought to play an important role in the response to salt and drought stresses. It has been determined that the expression levels of some genes are different between different cultivars (Elkoca-05 and Serra) of the same species. This study examines the role of β -Gal genes in drought and salt stress for the first time and serves as a preliminary study for future stress and functional studies.

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Author contribution Ayşe Gül Kasapoğlu, Azize Buttanrı, Emre İlhan, and Murat Aydın conducted the experiment and wrote the paper; Azize Buttanrı, Selman Muslu, Ayşe Gül Kasapoğlu, Burak Muhammed Öner, Ahmed Sidar Aygören, and Ertan Yıldırım participated in the bioinformatic analysis and stress applications; all the authors participated in the discussion of the article.

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Data Availability Data supporting the findings of this study are available from the corresponding author [PM] on request.

Declarations

Conflict of interest The authors have no conflicts of interest to declare.

Ethics approval and consent to participate This research study complies with research publishing ethics. In addition, the manuscript meets all applicable standards with regard to the ethics of experimentation and research, and there is no duplicate publication, fraud, plagiarism, or concerns about animal or human experimentation.

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